THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which element conducts electricity but is a non-metal? 1

(a) bromine **(b)** sulphur
(c) graphite (carbon) **(d)** phosphorus

2. Aluminium foil is used to wrap food because it is: 1

(a) malleable and non-toxic **(b)** magnetic
(c) brittle **(d)** heavy

3. An aqueous solution of copper sulphate is: 1

(a) a heterogeneous mixture **(b)** an element
(c) a compound **(d)** a homogeneous mixture

4. The number of atoms in a molecule of H_2SO_4 is: 1

(a) 4 **(b)** 7 **(c)** 3 **(d)** 6

5. Silver tarnishes in air due to: 1

(a) reaction with sulphur compounds **(b)** evaporation
(c) melting **(d)** reaction with nitrogen

6. Which can be separated by a magnet? 1

(a) sugar and water **(b)** salt and sugar
(c) iron sulphide **(d)** iron and sulphur mixture

7. The number of different elements in $NaHCO_3$ is: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

8. Which of the following will NOT rust? 1

(a) iron rod **(b)** stainless steel **(c)** iron nail **(d)** mild steel

9. Which among the following is a compound? 1

(a) iron **(b)** oxygen **(c)** baking soda **(d)** graphite

10. The total number of atoms in $C_6H_{12}O_6$ is: 1

(a) 12 **(b)** 6 **(c)** 18 **(d)** 24

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Is milk a mixture or a compound? Give reasons. 2

12. Why are copper and aluminium used in electric wires? 2

13. Why are metals good conductors of heat and electricity? 2

14. Why do metals produce a ringing sound? Give two uses of this property. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Classify as element, compound or mixture: sea water, copper, baking soda, brass, oxygen, ammonia. 3

16. Describe how electrolysis shows that water is a compound. 3

17. Explain why carbon is called a unique element with the example of diamond and graphite. 3

OR

Explain how an alloy is different from a compound. Use brass and water as examples. 3

18. Why is sugar solution considered a homogeneous mixture but a muddy water a heterogeneous mixture? 3

OR

Explain the importance of knowing the properties of elements in choosing materials. Give three examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe five physical properties of metals and compare them with non-metals. Give two exceptions. 5

OR

Describe how the concept of mixture helps in the daily life tasks of cooking, cleaning and water purification. 5

20. A mixture contains sand, iron filings, common salt and camphor. Describe a scheme of separation of all four. 5

OR

Explain the nature, properties and uses of noble gases and halogens (chlorine) with examples. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND